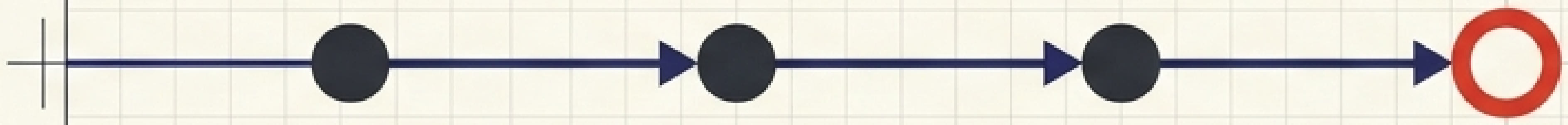


Learn to Reason About History.

A Seven-Volume Curriculum on Git as an Object Model, an Evidentiary Graph, and an Architecture for Agents.

Flyxion | Independent Researcher | 2026

Don’t Memorize Commands. Reason About History.

The Rote Paradigm

Goal: Memorize commands against imagined scenarios.

Mental Model: A folder with a magic save button.

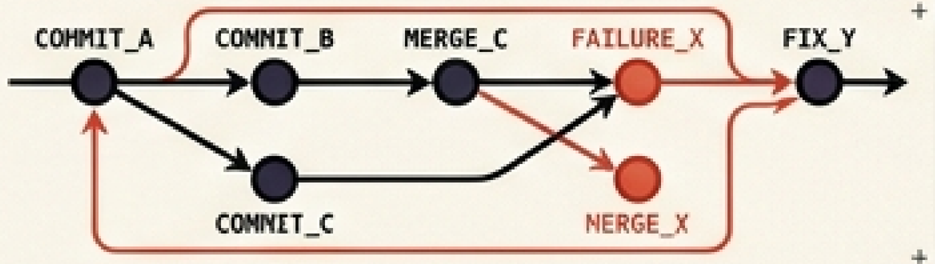
The Textbook: Independent text offering contrived examples.

The History: Sanitized, linear, clean (a theoretical lie).

The History Paradigm

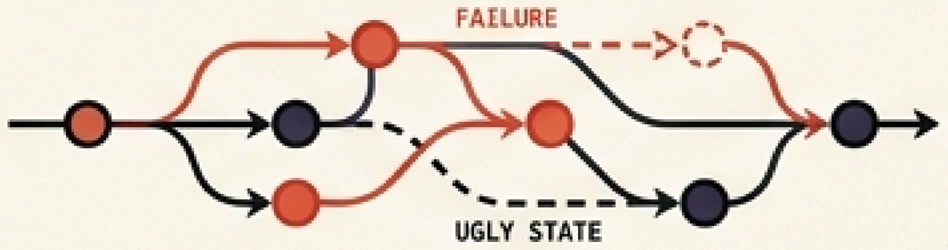
Goal: Predict the content-addressed graph before executing.

Mental Model: A directed acyclic graph (DAG) of immutable evidence.

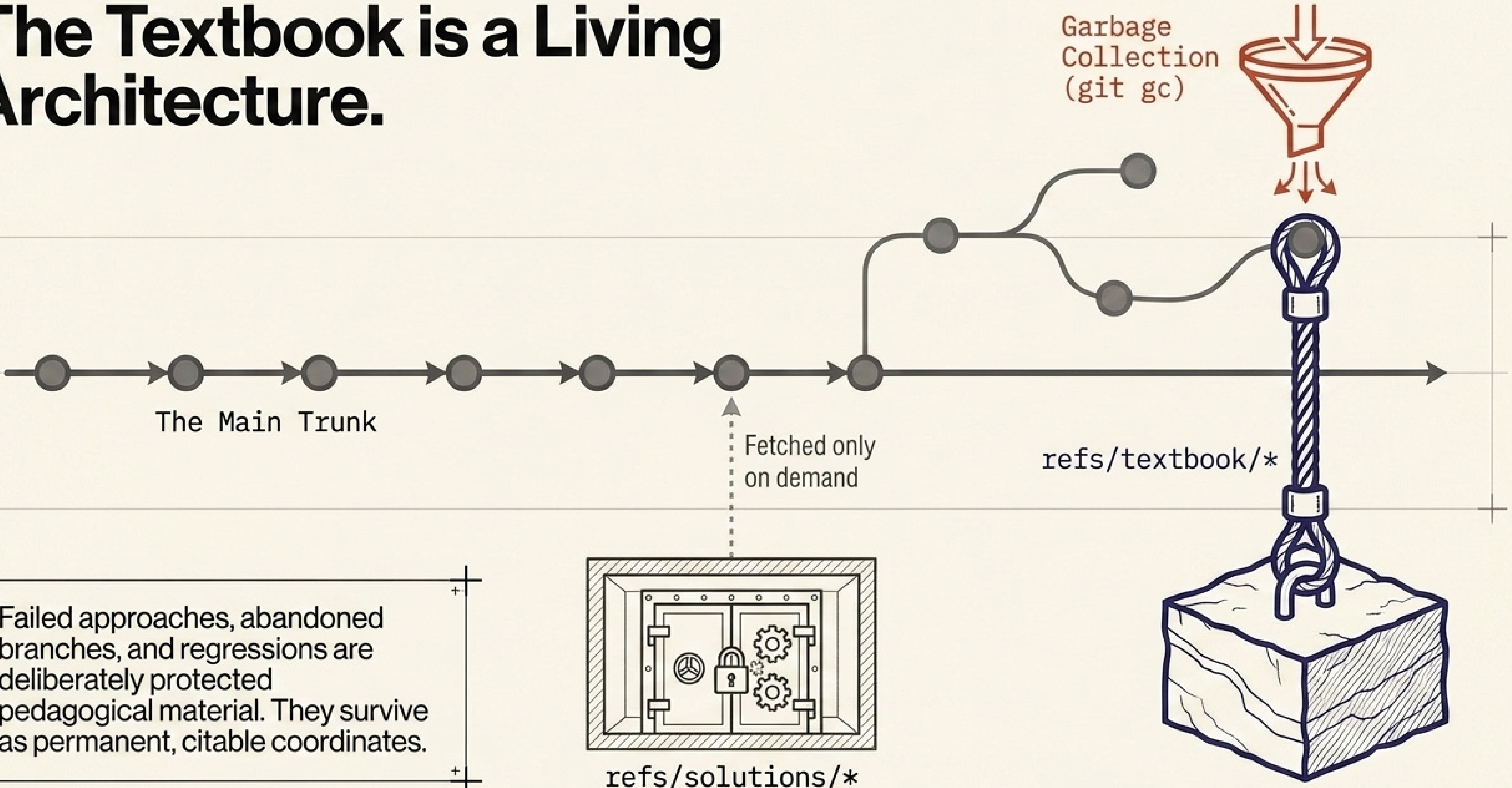


The Textbook: The repository is the textbook.

The History: Preserved failures, ugly intermediate states, authentic topology.



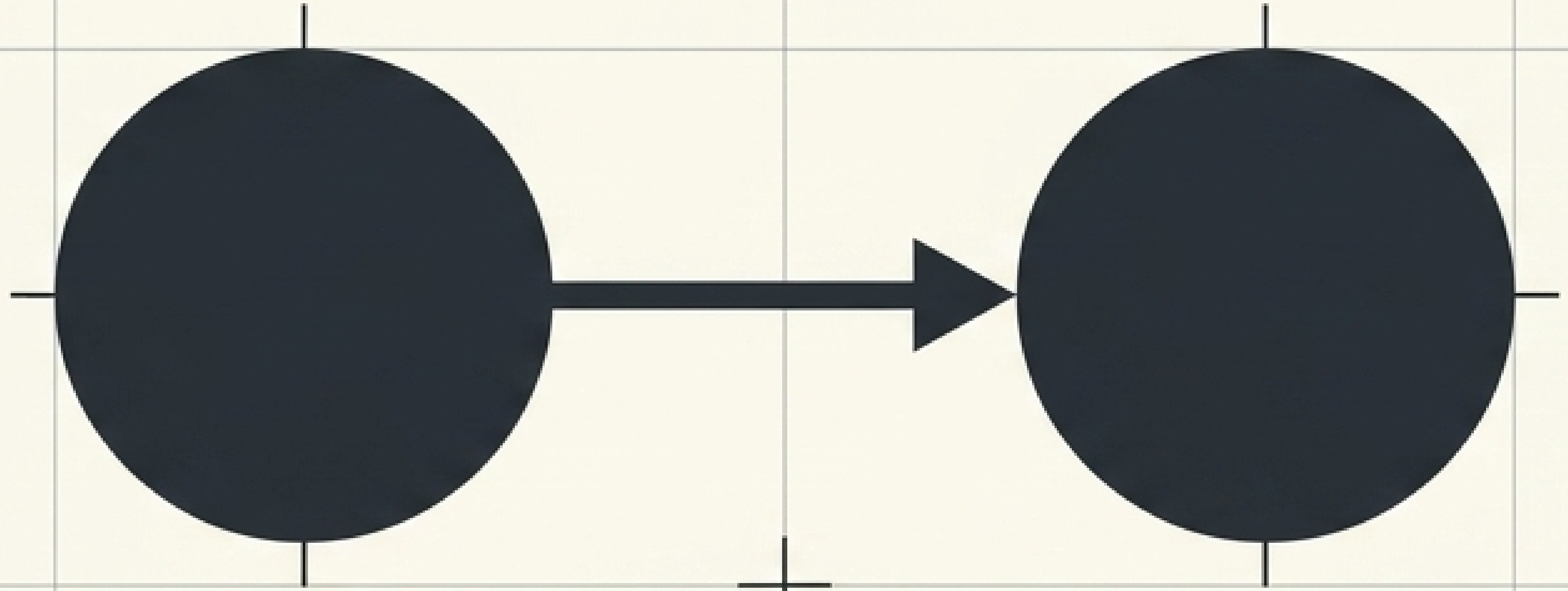
The Textbook is a Living Architecture.



Failed approaches, abandoned branches, and regressions are deliberately protected pedagogical material. They survive as permanent, citable coordinates.

Volume I: Git from First Principles.

[APP: Dockerized Personal Web Service]



$$S_i \rightarrow S_{i+1}$$

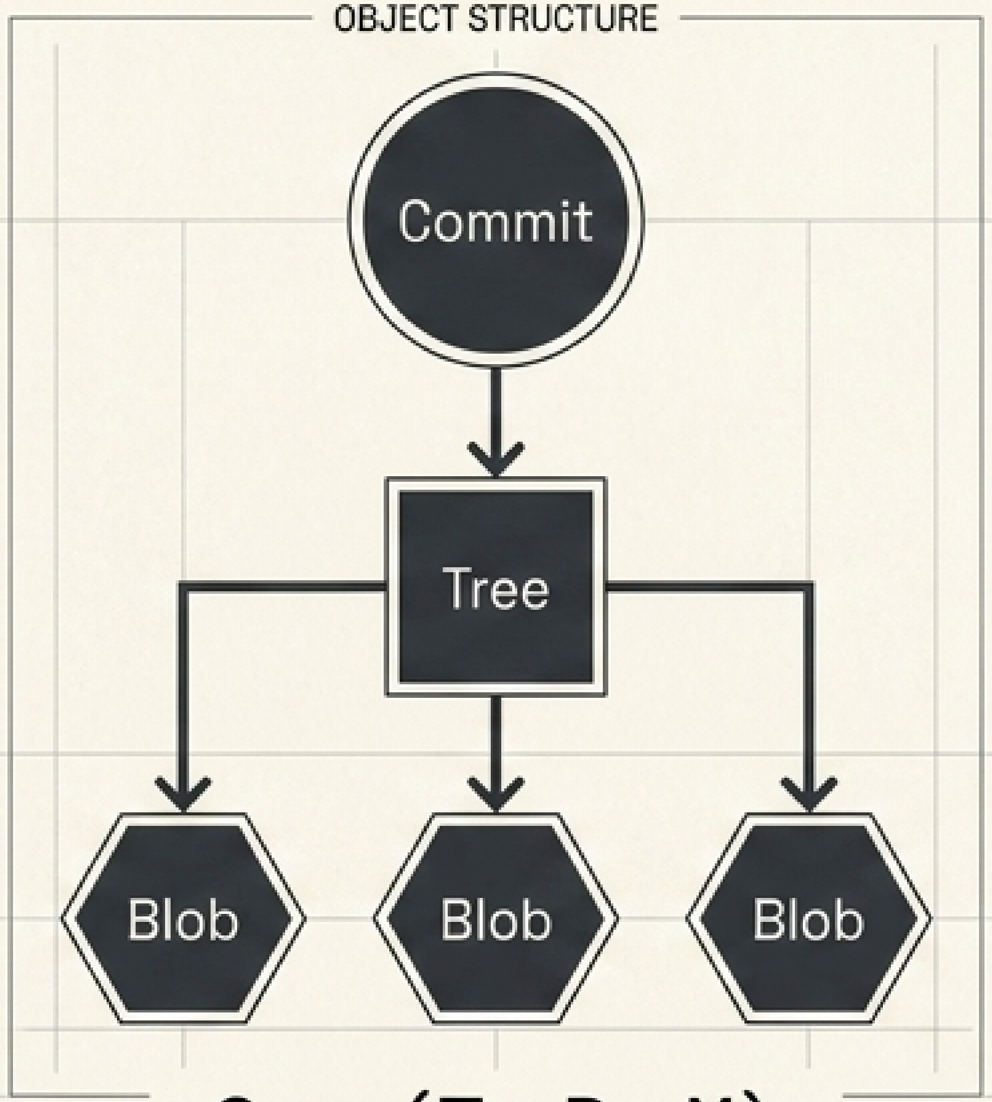
THE COMMIT AS A TRANSITION.

The reader begins with almost nothing: a directory whose state matters. Commands appear only once the conceptual machinery demands them.

Change › Remember › Compare › Name › Diverge › Combine › Recover › Publish

Volume II: Inside Git.

[APP: Versioned Stable Diffusion Workstation]



C = (T, P, M)

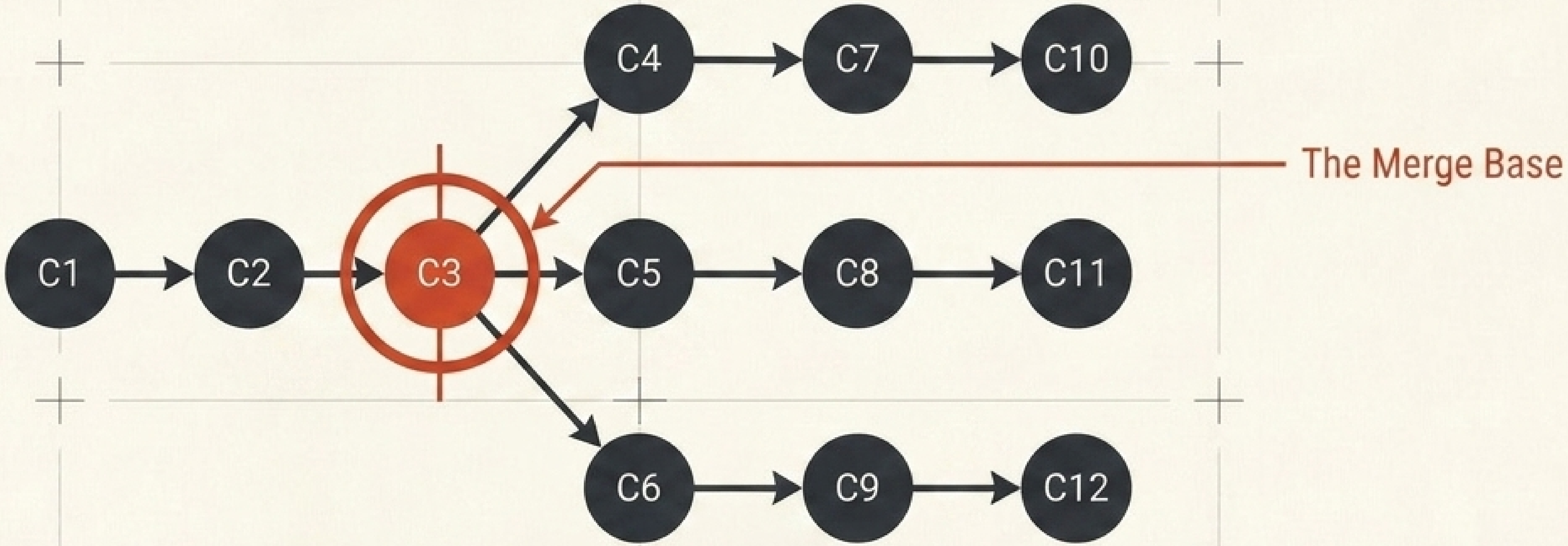
THE COMMIT AS AN OBJECT.

A multi-gigabyte checkpoint forces a distinction between logical identity and physical storage. We introduce a model manifest—Git retains only an identity pointer, not the 7GB artifact.

File ⟶ Content ⟶ Object ⟶ Tree ⟶ Commit ⟶ Reference ⟶ Reachability

Volume III: The Geometry of Git.

[APP: Ollama/Granite Research Assistant]



$$C_i \in V(G)$$

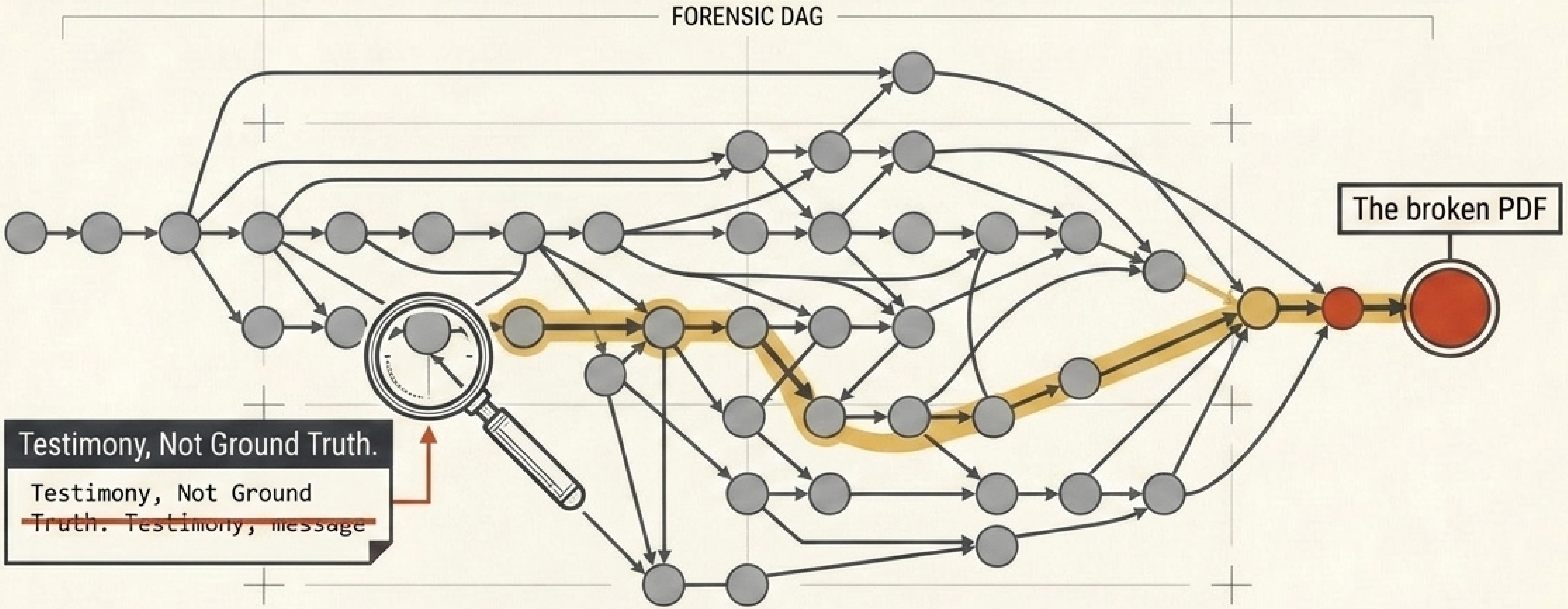
THE COMMIT AS A GRAPH VERTEX.

Branches transform from an invented teaching device into the literal historical representation of genuine competing hypotheses. Ancestry determines dependence, not the wall-clock time.

Ancestry ⟶ Divergence ⟶ Comparison ⟶ Merge Base ⟶ Integration ⟶ Transplantation ⟶ Reconstruction

Volume IV: Git Archaeology.

[APP: Essay-Writing and LaTeX Pipeline]



THE COMMIT AS EVIDENCE.

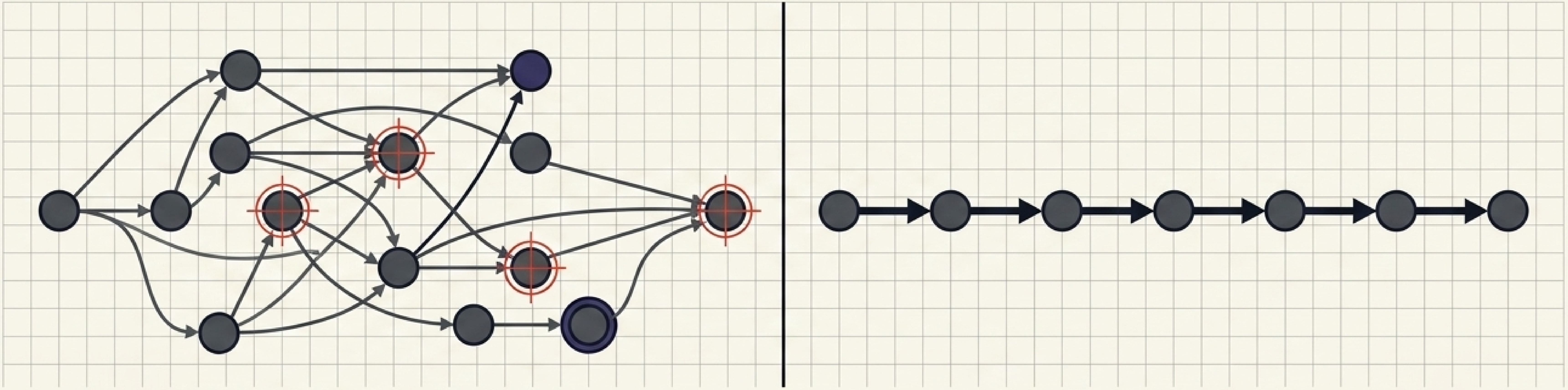
Git preserves evidence without automatically explaining what the evidence means. A textual modification does not equal a semantic modification.

Symptom → Evidence → Localization → Attribution → Causation → Recovery → Reconstruction

Volume V: Git in the Open

[APP: Open-Source Scientific-Computing Library]

The Social Consequences Model



Authentic Topology (Merge)

Synthesized Linear Lie (Rebase)

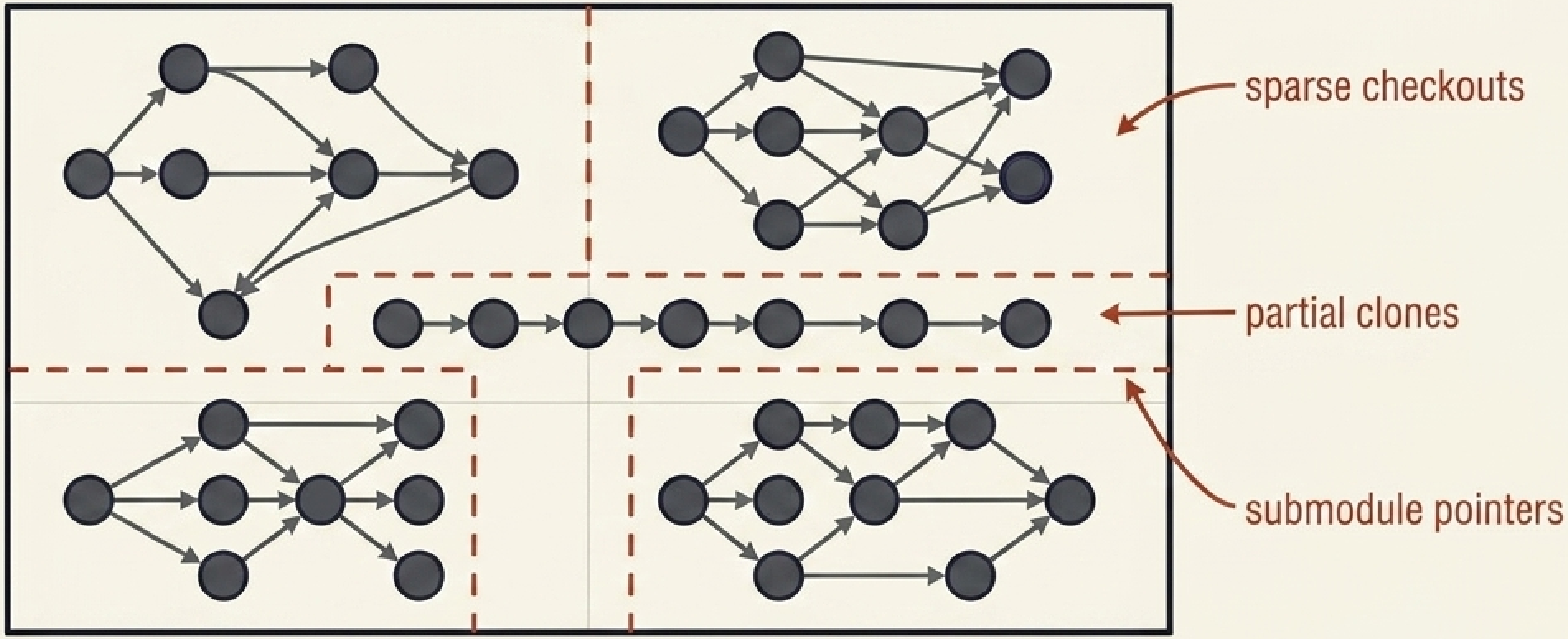
THE COMMIT AS A UNIT OF EXCHANGE

Same code, different histories. Identical trees integrated differently have profound consequences for archaeology, bisectability, and attribution.

Repository → Remote → Exchange → Collaboration → Review → Integration → Governance

Volume VI: Git at Scale.

[APP: The Reconnected System (Monorepo)]



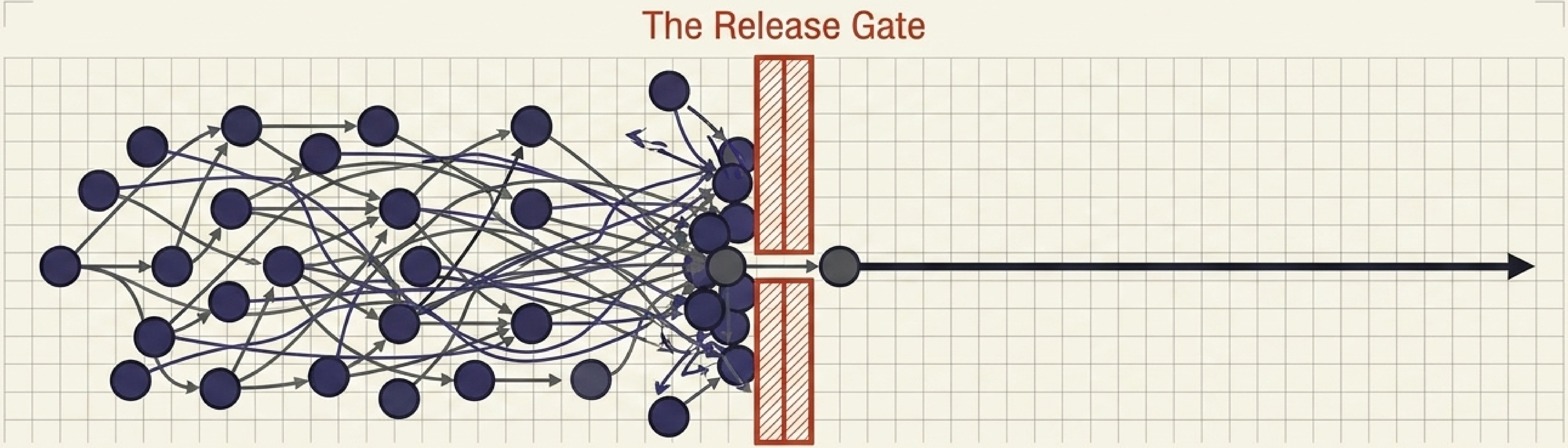
THE COMMIT AS AN ARCHITECTURAL BOUNDARY.

Whether a commit belongs in this history or another is no longer a question the commit itself can answer, only one the repository's design can.

Project ⟶ Boundary ⟶ Workspace ⟶ Dependency ⟶ Artifact ⟶ Automation ⟶ Scale

Volume VII: Git for Agents.

[APP: The Inherited Delegated System]

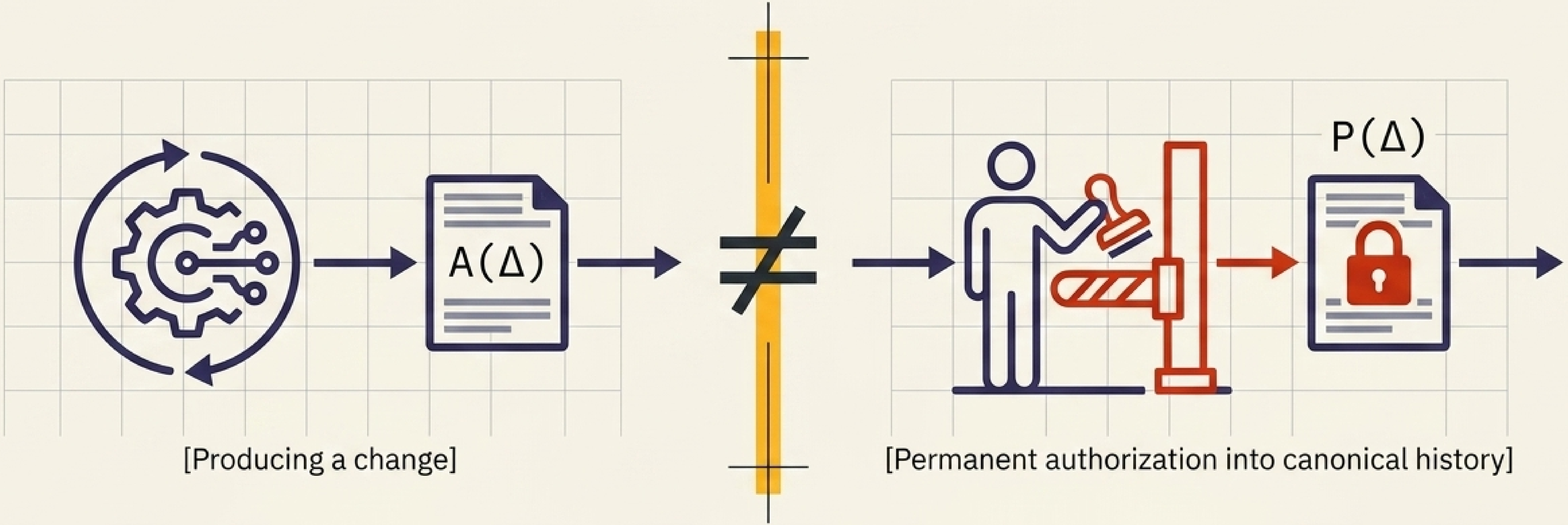


THE COMMIT AS A COMPRESSION BOUNDARY.

Machines can write faster than humans can read. When generation becomes cheap, the scarce resources become attention, verification, and coherent history.

Task → Scope → Steering → Interruption → Verification → Provenance → Integration

The Delegation Theorem.



$$A(\Delta) \neq P(\Delta)$$

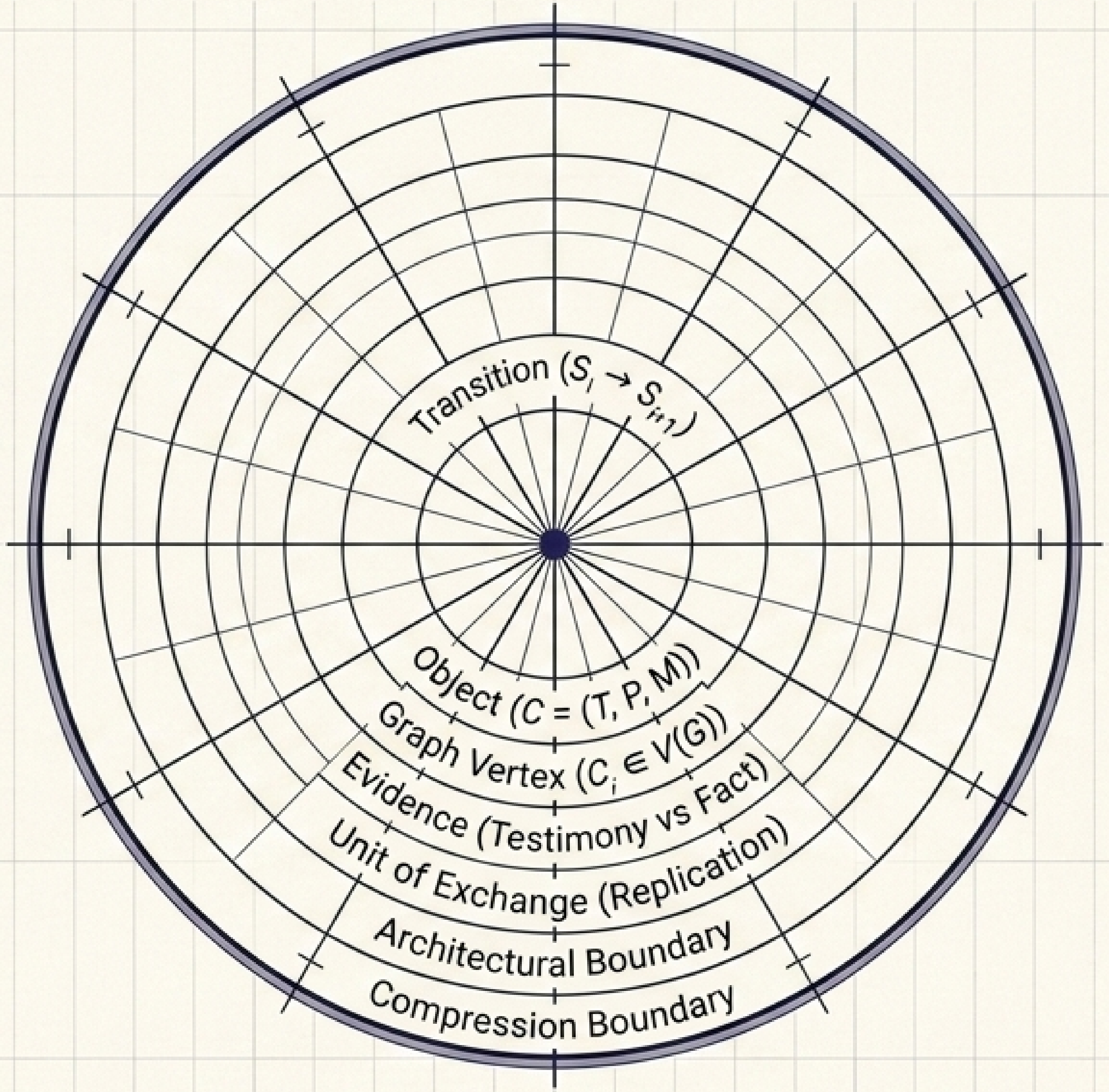
Producing a transformation does not, by itself, authorize that transformation to enter canonical history. Capability ≠ Authorization.

The Symmetries of History

The Graph (Micro Scale)	Volume III (Forward Design) Target History → Git Operations. (Building the graph).	Volume IV (Backward Reconstruction) Broken Graph → Possible Causes. (Interrogating the graph).
The Boundaries (Macro Scale)	Volume VI (Architecture) Where should the boundaries of history lie? (Defining the walls).	Volume VII (Delegation) Who may alter history within those boundaries? (Building inside the walls).

The underlying skill—reasoning correctly about what an operation does to a graph—is identical in both directions.

The Evolution of the Commit.



RECORD → REPRESENT → TRANSFORM → INFER → COORDINATE → ARCHITECT → DELEGATE.

The Ending Was Always True.

Agents do not introduce
$A(\Delta) \neq P(\Delta)$ into Git.
They merely make it
it impossible to keep ignoring.

Even Volume I's very first
exercise—inspecting `git diff`
before typing `git commit`—taught
exact separation in miniature.

**Generation may be autonomous.
Commitment should remain deliberate.**

`// A commit is a commitment.`